
EN2091 - Laboratory Practice and Projects

Photoelectric Smoke Detector

Project Proposal

Group Name: Perceptrons

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1 Introduction

Smoke detectors are used in both industrial and residential environments to mitigate fire risk. There are two commonly used types; radioactive smoke detectors and photoelectric smoke detectors. Photoelectric models are best suited for smouldering fires [1].

2 Functionality

The key functionality of a smoke detector is to trigger a fire alarm in the presence of smoke. In a photoelectric smoke detector, this is achieved through an infra-red emitter and a receiver housed in a specially designed chamber open to the environment. The infra-red beam does not directly fall on the receiver as per their alignment. However, when smoke has entered the chamber, the infra-red beam is scattered, and therefore the receiver can sense the scattered light. This signal is used to make the decision of triggering the alarm appropriately. The alarm will go off [2].

The device will have a built in audio alarm, and means to indicate that it is operating properly. Further, the alarm mechanism is capable of handling hysteresis and false positives. It will be powered by a non-rechargeable battery and optimized for low-power operation.

3 System Block Diagram

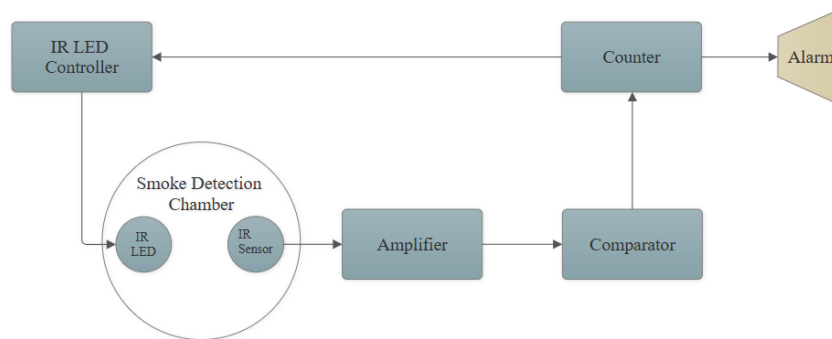


Figure 1: System Block Diagram

4 Methodology Overview

The device will contain infra-red LED and a photo-diode housed in a special chamber. This chamber will be designed such that the IR beam does not fall on the photo-diode, and the light coming from outside is restricted. However, air circulation will be allowed between the chamber and the environment. The IR LED is lit as pulses to minimize power consumption. In case of a fire, smoke will enter the chamber and cause the IR beam to scatter, resulting an output from the photo-diode. This signal will be amplified and fed into a comparator. If the signal is beyond a calibrated threshold, the pulse frequency will increase. After a certain number of consecutive pulses with photo-diode output, the alarm will be triggered. After a false-positive, the counter will reset once the photo-diode output turns low, and the alarm will not be triggered [3].

5 Micro-products and Interconnections

5.1 IR LED Controller

This module contains an oscillator with a square pulse output, which can switch between two frequencies depending on the state of the counter. The default frequency is lower and it will switch to the higher frequency if the counter is not at its initial state. This pulse waveform is given to the IR-LED and as the clock to the counter as well.

5.2 Smoke Detection Chamber

This is specially designed as per the previously mentioned requirements and houses the IR LED and the photo-diode.

5.3 Amplifier

This block amplifies the output of the photo-diode.

5.4 Comparator

The comparator compares the amplified output with a calibrated threshold, handling hysteresis as well. This module stands between the analog and digital domains.

5.5 Counter

This is the only digital block in the system. Once a photo-diode output is detected for a pulse, the counter starts counting such pulses. If any pulse came without a photo-diode output, the counter will be pulled to the initial (0) state. Once a certain number of pulses are counted, the counter will trigger the alarm. Therefore, the counter module act as the central control of the system.

5.6 Interconnections

All the micro-products will be directly implemented on the same printed circuit board interconnected with PCB traces. The smoke detection chamber will also be fastened to the circuit board. Power will be supplied through a non-rechargeable battery.

6 Micro-product allocation among group members

- **Maduwage P. K.** - Detection Chamber, Counter module
- **Herath M. H. M. S. N.** - PCB Design, Counter module
- **Harsiyan S. A.** - PCB Design, IR LED Controller module
- **Himash B. C.** - Enclosure, Amplifier and Comparator modules

References

- [1] L. Tyler, “What is a photoelectric smoke detector?” May 2024. [Online]. Available: <https://internationalfireandsafetyjournal.com/photoelectric-smoke-detector/>
- [2] T. Verbeure, “Taking apart a smoke detector,” April 2020. [Online]. Available: <https://tomverbeure.github.io/2020/04/10/Smoke-Detector.html>
- [3] Microchip Technology Inc., *RE46C190: CMOS Low Voltage Photoelectric Smoke Detector ASIC with Interconnect and Timer Mode Data Sheet*, Microchip Technology Inc. [Online]. Available: <https://ww1.microchip.com/downloads/en/DeviceDoc/RE46C190-DS-20002271C-Final1.pdf>